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Box-enhanced context fusion for citation intent classification

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ABSTRACT

Citation intent classification is a core research topic in scientometrics and information retrieval. This task analyzes citation contexts and relationships to categorize citation behaviors such as providing support, making comparisons, or offering contextual background. Identifying these behaviors helps reveal the specific role of citations in scholarly communication and clarifies the underlying authorial intent. Existing text-encoding approaches typically formulate citation intent classification as a conventional text classification task. However, this formulation overlooks two critical factors: the influence of citation marker position on classification performance and the implicit hierarchical relationships among texts within the same class. To address these limitations, we propose GLB-Cite, a novel citation intent classification model that fuses local citation marker information with a global context representation to obtain a unified fused representation. This representation is then projected into two parallel spaces: a vector embedding space for semantic discrimination and a box embedding space for modeling overlap between citation contexts and intent classes. By jointly optimizing these two branches, GLB-Cite balances semantic expressiveness and geometric boundary modeling, yielding notable gains especially for intent classes with overlapping semantics and ambiguous decision boundaries. We evaluate the performance of GLB-Cite on two public datasets, SciCite and ACL-ARC, using the macro-F1 score as the primary evaluation metric. Experimental results show that GLB-Cite achieves state-of-the-art performance, with F1 scores of 87.69% on SciCite and 79.79% on ACL-ARC.

1. Introduction

Citation intent classification (CIC) is an important research topic at the intersection of bibliometrics, scientometrics, and natural language processing (NLP). Early work introduced annotation schemes for citation function classification, highlighting the importance of systematically characterizing the role of citations in scholarly discourse (Teufel et al., 2006). Within NLP research, citation intent classification is commonly studied as the task of classifying the semantic role or function of citation statements (Jurgens et al., 2018; Teufel et al., 2006). Recent work has further argued that citation context analysis should move beyond the single-sentence, single-label setting in order to better capture the complexity of real citation behavior (Lauscher et al., 2021). Current classification systems found in ACL, ACM, and IEEE studies typically employ six intent classes, although simplified three-class taxonomies remain prevalent.

CIC not only helps reveal the underlying logic of scientific development, but also supports practical applications such as literature retrieval, research evaluation, and academic recommendation systems (Zhu et al., 2015). Advances in AI are further amplifying its significance, position-

ing CIC as a pivotal research area in scientometrics and information retrieval. Academically, CIC enables deeper insights into scientific communication patterns by categorizing citations, thereby tracing theoretical evolution, pinpointing disciplinary controversies, and mapping interdisciplinary knowledge flows (Small, 2011). Highly cited studies, for instance, might reflect contentious debates rather than general consensus (Greenberg, 2009). Such fine-grained analysis facilitates the construction of more precise knowledge networks and further advances scientometric research.

At the applied level, CIC addresses critical limitations in current academic systems. Existing bibliometric systems and citation-based analyses often fail to capture the nuanced roles of individual citations (Zhang et al., 2025). Integrating CIC allows researchers to efficiently identify seminal works, easily distinguishing between foundational studies that support a theory and disruptive research that challenges established norms. Moreover, citation-based evaluation should take into account that not all citations carry equal academic influence (Zhu et al., 2015). CIC mitigates this by introducing intent-aware evaluations, such as distinguishing affirming from critical citations, enabling fairer assessments of scholarly impact (Jurgens et al., 2018).

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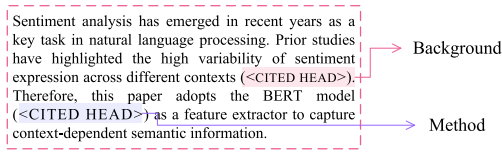


Fig. 1. The importance of citation marker position information.

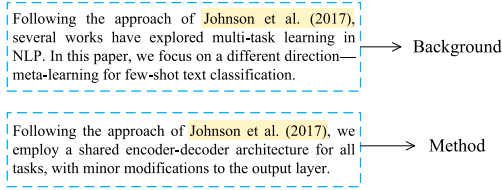


Fig. 2. The importance of the global context representation surrounding the citation.

Despite recent advances, current state-of-the-art citation intent classification models achieve an F1 score of 86.84% for 3-class classification but only 76.50% for the more challenging 6-class task, indicating significant room for improvement (Shui et al., 2024). Recent advances in artificial intelligence, especially in natural language processing and box embedding methods, have created new opportunities to improve citation intent classification.

(1) Citation marker position awareness and context fusion modeling

Our analysis reveals that citation intent classification differs from conventional text classification in that it is highly sensitive to the citation marker position in the input sequence. As demonstrated in Fig. 1, citation markers appearing at different positions within the same sentence can convey different intents. Inspired by (Shui et al., 2024), we introduce citation marker position embedding to strengthen the modeling of local information around the citation marker.

However, focusing only on the local representation around the citation marker is insufficient. As illustrated in Fig. 2, changes in the surrounding context can substantially alter citation intent even when the citation marker appears in the same position. We therefore complement local information with a global context representation to better distinguish citation intents.

(2) Modeling class boundary overlap with box embedding

Our analysis reveals significant inter-class confusion in boundary regions, as evidenced by the t-SNE visualization in Fig. 3. This issue stems from the limitations of traditional vector-based approaches in modeling boundary ambiguity among citation intents. Intuitively, vector embeddings represent a citation as a point in semantic space, whereas a citation intent often behaves more like a bounded semantic region. For example, a citation describing a methodology may overlap with concepts such as data usage or experimental setup. Point-based embeddings (Vilnis & McCallum, 2014) are not well suited to modeling such containment and intersection relationships. To address this limitation, we introduce box embeddings (Lv et al., 2024; Onoe et al., 2021), which represent concepts as hyper-rectangles rather than points. In this formulation, overlap naturally reflects semantic ambiguity, while containment reflects inclusion-like relationships. This property makes box embeddings well suited to handling boundary ambiguity in citation intent classification.

(3) Unified Semantic-Geometric Fusion Framework

Vector embeddings effectively capture fine-grained intra-class semantic variation, whereas box embeddings are better suited to modeling inter-class boundary overlap. Motivated by these complementary strengths, we propose a joint framework that improves both semantic discrimination and geometric boundary modeling. Specifically, the fused representation is optimized through two parallel branches, one in vector space and the other in box space. This joint optimization encourages intra-class compactness in vector space while improving inter-class

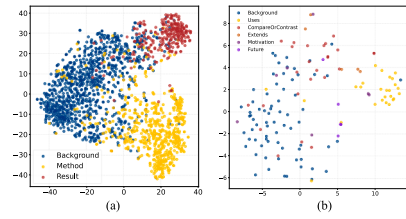


Fig. 3. t-SNE visualization results on the SciCite and ACL-ARC datasets.

separation in box space, thereby alleviating boundary ambiguity in citation intent classification.

To this end, we propose **GLB-Cite**, a **dual-embedding citation intent classifier that jointly models semantic and geometric spaces**. Our approach specifically tackles: (1) limited contextual semantics, (2) ambiguous class boundaries, and (3) constrained embedding expressiveness in existing methods. Key contributions include:

- 1) For citation context modelling, we propose **GL-Fusion**. This context encoding module fuses the local representation with the global representation to more adequately capture both local and global information in citation intent classification.
- 2) For representation space design, we introduce **box embeddings** into citation intent classification to model overlapping class regions and inclusion-like relations between citation contexts and intent classes. This geometric design is especially useful for classes with ambiguous semantic boundaries, and we further verify its class-dependent effect through detailed ablation analysis.
- 3) For optimization, we propose a **joint training strategy** that combines the vector-space loss and the box-space loss, thereby improving semantic discrimination and geometric boundary modeling simultaneously.

The remainder of this work is organized as follows: Section 2 reviews prior research on citation intent classification and the development of box embedding techniques. Section 3 details our proposed GLB-Cite framework, including: (1) task formulation, (2) system architecture, (3) local-global feature fusion, (4) box-based geometric modeling, and (5) the joint optimization strategy. Section 4 presents experimental results through comparative benchmarks, ablation studies, and visual analysis, with comprehensive performance evaluation.

2. Related work

2.1. Citation intent classification

Prior research has explored various approaches to citation intent classification. Early work proposed annotation schemes for citation function classification (Teufel et al., 2006). (Cohan et al., 2019) proposed a multi-task framework combining chapter title prediction and citation necessity detection to enhance classification performance. The MTCIC model (Qi et al., 2023) further incorporated two auxiliary binary classification tasks: citation worthiness and key citation identification. To address class imbalance, EnsIntWS (Paolini et al., 2024) employed ensemble learning through binary classification of classes. Graph-based methods like GraphCite (Berrebbi et al., 2022) introduced heterogeneous citation graphs incorporating paper, author, and venue information, while CitePrompt (Lahiri et al., 2023) adapted prompt learning techniques for this task.

However, these approaches share a critical limitation: they overlook the positional sensitivity inherent to the citation marker, a key distinction from general text classification. Although MTL (Shui et al., 2024) attempted to model dataset relationships through task-relationship learning (TRL), its readout function over-emphasizes citation marker positions while neglecting broader contextual information.

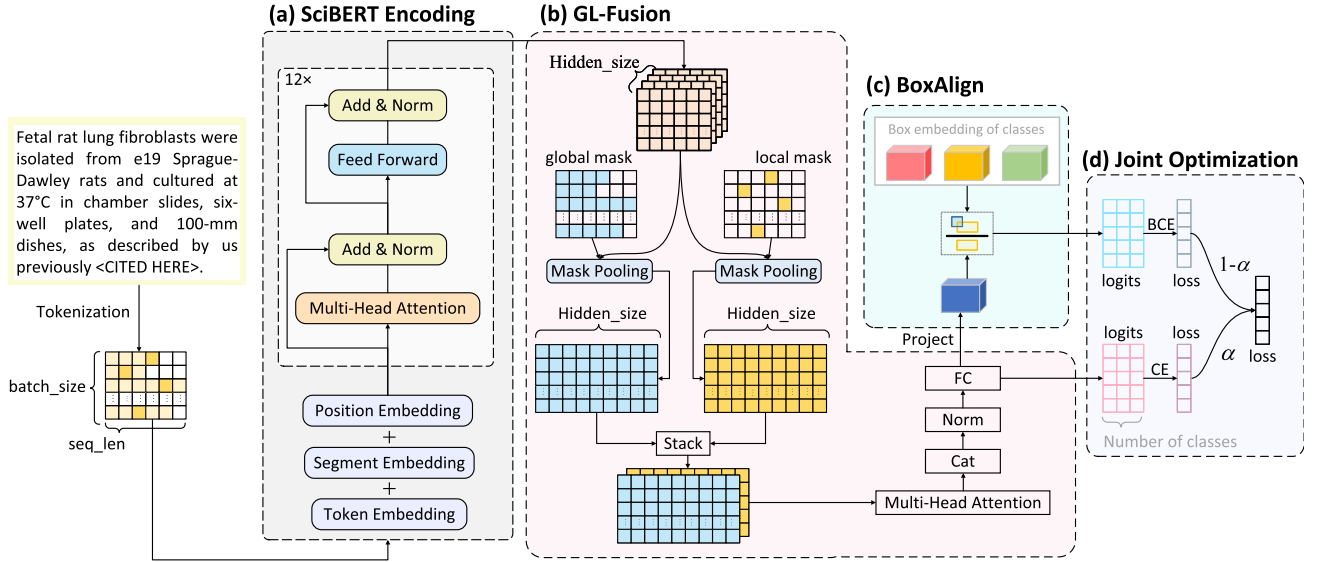


Fig. 4. The overall architecture of GLB-Cite.

Recent work by (Koloveas et al., 2025) evaluated various large language models (LLMs) - including LLaMA (Grattafiori et al., 2024), Mistral (Jiang et al., 2023), Phi (Abdin et al., 2024), Gemma (Team et al., 2024), and Qwen (Team, 2024) - for citation intent classification using context learning methods under different fine-tuning instructions. Their results revealed two key limitations: (1) smaller LLMs (1B-3B parameters) failed to achieve satisfactory performance, while (2) fine-tuning larger LLMs (7B+ parameters) required substantial computational resources and time yet only matched SciBERT's performance. Meanwhile, ICCL (Liu et al., 2024) introduced an in-context curriculum learning strategy that orders demonstrations by difficulty, which can be assessed by human judgment or model-based metrics such as perplexity. Despite these efforts, current LLMs still underperform SciBERT-based approaches in citation intent classification tasks.

2.2. Box embedding

Traditional text representation methods predominantly utilize vector embedding, where each linguistic unit (e.g., word, sentence, or class) is mapped to a single vector in Euclidean space, as implemented in Word2Vec (Mikolov et al., 2013), GloVe (Pennington et al., 2014), and Transformer-based architectures like BERT (Devlin et al., 2019) and SciBERT (Beltagy et al., 2019). While these approaches effectively leverage vector distance metrics (e.g., cosine similarity, dot product) for semantic analysis and achieve strong performance across various NLP tasks, they exhibit inherent limitations in capturing structured semantic relationships. Specifically, the absence of explicit geometric boundaries in point-based representations hinders their ability to model: (1) hierarchical structures, (2) set inclusion relationships, and (3) ambiguous class boundaries where semantic overlap occurs.

To address these structural modeling limitations, box embeddings (Vilnis et al., 2018) represent semantic objects as axis-aligned hyper-rectangles defined by minimum and maximum boundary vectors.

This geometric formulation provides three key advantages over vector embedding: (1) explicit spatial extent enables precise boundary representation, (2) native support for set operations (containment, intersection, overlap), and (3) inherent capacity to model hierarchical relationships. These properties make box embeddings particularly effective for handling fuzzy boundaries and multi-attribution scenarios in semantic spaces.

Recent research has increasingly adopted box embeddings for their ability to model complex semantic relationships in knowledge graphs

and classification tasks. (Onoe et al., 2021) demonstrated their superiority over vector representations in entity type classification, particularly for hierarchical relationships and predictive calibration. For knowledge graphs, BEURRE model (Chen et al., 2021) leverages box embeddings to enable uncertainty-aware reasoning through higher-order dependency modeling, while Concept2Box (Huang et al., 2023) combines box and vector representations for enhanced hierarchical and instance feature modeling. Practical applications have also emerged, with box-based recommender systems like InBox (Xu et al., 2024) and BoxMAKR (Liang et al., 2024) effectively capturing user interest ranges.

In summary, box embeddings provide a principled geometric framework for modeling three fundamental relationships: (1) subset containment, (2) semantic overlap, and (3) indeterminate boundaries. By leveraging explicit spatial boundaries, they effectively address the inherent limitations of vector embedding in representing hierarchical semantic structures. This capability offers particularly valuable insights for citation intent classification, where boundary ambiguity and class overlap present persistent challenges.

3. Method

3.1. Problem definition

Let the citation intent classification dataset be denoted by the symbol $D^I = \{(x_i^I, y_i^I)\}_{i=1}^N$ where $N = |D^I|$, with $x_i^I \in \mathcal{X}^I$ representing the textual context surrounding the citation marker <CITED HEAD>, and $y_i^I \in \mathcal{Y}^I$ being the citation intent label. Let $K = |\mathcal{Y}^I|$ be the number of classes, then $\mathcal{Y}^I = \{0, 1, \dots, K-1\}$. For any class $c \in \mathcal{Y}^I$, we define $n_c^I = |\{(x_i^I, y_i^I) \in D^I \mid y_i^I = c\}|$ as the number of samples in class c . Our goal is to train a classifier $C: \mathcal{X}^I \rightarrow \mathcal{Y}^I$ that optimizes the objective in Eq. (1).

$$C(x_i^I) = y_i^I, \quad \forall (x_i^I, y_i^I) \in D^I. \quad (1)$$

3.2. Framework

Our proposed model, **GLB-Cite** (illustrated in Fig. 4), integrates multi-granular contextual information and uses dual representation spaces to enhance class discrimination. The framework consists of four modules: SciBERT Encoding, GL-Fusion (Global-Local Fusion), BoxAlign (Geometric Alignment), and Joint Optimization.

Fig. 4 illustrates the overall pipeline, which consists of context fusion and dual-space representation learning. First, SciBERT encodes the

input text, and GL-Fusion integrates a local representation centered on the citation marker with a global representation aggregated from the full sentence context, yielding a fused representation. Next, this fused representation is projected into two parallel spaces: a vector embedding space for semantic discrimination and a box embedding space, implemented by BoxAlign, for modeling geometric boundaries and overlap relationships. Finally, the two branches are trained jointly through a learnable weighting coefficient α , which is obtained by applying a sigmoid function to a trainable scalar parameter.

For clarity, we use the following terminology consistently throughout this paper. The *local representation* denotes the representation centered on the citation marker position, whereas the *global representation* denotes the representation aggregated from the context of the entire sentence. Their integration forms the *fused representation*. This fused representation is then projected into two parallel spaces: a *vector embedding space* for semantic discrimination and a *box embedding space* for geometric boundary modeling.

3.3. SciBERT encoding

The token sequence of the i th citation context is defined in Eq. (2).

$$x_i^t = [\text{CLS}], x_{i1}^t, x_{i2}^t, \dots, x_{iN}^t, [\text{SEP}], \quad (2)$$

here, x_{ij}^t represents the j -th token, and N denotes the total number of tokens in the text x_i^t , excluding the special [CLS] and [SEP] tokens.

The contextual embedding matrix produced by the pre-trained SciBERT encoder is defined in Eq. (3).

$$h_i^t = \text{SciBERT}(x_i^t) \in \mathbb{R}^{(N+2) \times d}. \quad (3)$$

Here, SciBERT serves as the text encoder that maps citation contexts into a latent feature space, while a multilayer perceptron (MLP) is used to produce the final class predictions.

3.4. Feature fusion

The model extracts two types of representations: (1) a global representation from the full citation context and (2) a local representation from the citation marker position. The GL-Fusion module integrates these two representations, as shown in Fig. 4(b).

To capture multi-granular semantics, we extract a local representation from the citation marker and a global representation from the full context using position-aware masking. Specifically, we define a local mask $\text{mask}_{i_{\text{ch}}}^t$ that focuses on the citation marker and a global mask $\text{mask}_{i_{\text{mean}}}^t$ that covers the entire sequence. The corresponding representations are derived as follows:

$$h_{i_{\text{mean}}}^t = \frac{1}{\sum_{j=1}^{N+2} \text{mask}_{i_{\text{mean}}}^t[j]} \sum_{j=1}^{N+2} \text{mask}_{i_{\text{mean}}}^t[j] \cdot h_i^t[j]. \quad (4)$$

$$h_{i_{\text{ch}}}^t = \frac{1}{\sum_{j=1}^{N+2} \text{mask}_{i_{\text{ch}}}^t[j]} \sum_{j=1}^{N+2} \text{mask}_{i_{\text{ch}}}^t[j] \cdot h_i^t[j]. \quad (5)$$

We propose an attention-based fusion module that learns semantic relationships between $h_{i_{\text{mean}}}^t$ (global) and $h_{i_{\text{ch}}}^t$ (local) representations. To ensure training stability and introduce non-linearity, the inputs are first transformed using a tanh activation function, as described in Eqs. (6) and (7), prior to self-attention processing.

$$h_{i_{\text{mean}}}^t = \tanh(h_{i_{\text{mean}}}^t), \quad (6)$$

$$h_{i_{\text{ch}}}^t = \tanh(h_{i_{\text{ch}}}^t). \quad (7)$$

The global and local representations, $h_{i_{\text{mean}}}^t$ and $h_{i_{\text{ch}}}^t$, are concatenated along the sequence dimension to construct an input sequence of length 2, detailed in Eq. (8).

$$H = [h_{i_{\text{mean}}}^t; h_{i_{\text{ch}}}^t] \in \mathbb{R}^{2 \times d}, \quad (8)$$

This sequence is then processed by multi-head self-attention to generate a context-aware fused representation, as defined in Eq. (9). Specifically, self-attention derives a 2×2 weight matrix that captures the semantic interaction between $h_{i_{\text{mean}}}^t$ and $h_{i_{\text{ch}}}^t$ and uses it to adaptively combine them.

$$H_{\text{attn}} = [h_{i_{\text{mean}}}^{t, \text{enhanced}}, h_{i_{\text{ch}}}^{t, \text{enhanced}}] = \text{MHSA}(H, H, H) \in \mathbb{R}^{2 \times d}. \quad (9)$$

The module applies multi-head self-attention to the sequence formed by $h_{i_{\text{mean}}}^t$ and $h_{i_{\text{ch}}}^t$, thereby modeling their semantic interaction and producing enhanced representations, which are then concatenated into a single fused representation, as defined in Eq. (10). More specifically, attention computes a 2×2 weight matrix through scaled dot-product attention followed by softmax, enabling adaptive combination of global and local information.

$$h_{i_{\text{fusion}}}^t = h_{i_{\text{mean}}}^{t, \text{enhanced}} \parallel h_{i_{\text{ch}}}^{t, \text{enhanced}} \in \mathbb{R}^{2d}, \quad (10)$$

We apply layer normalization to $h_{i_{\text{fusion}}}^t$ to maintain stability, and then project this normalized output into the final representation space using a fully connected network, as expressed in Eq. (11).

$$h_{i_{\text{fusion}}}^t = \mathbf{W} \cdot \text{LayerNorm}(h_{i_{\text{fusion}}}^t) + \mathbf{b}, \quad (11)$$

where $\mathbf{W} \in \mathbb{R}^{2d \times 2d}$ is the learnable parameter and $h_{i_{\text{fusion}}}^t \in \mathbb{R}^{2d}$ denotes the final fused representation.

3.5. Box enhancement

To explicitly model the containment and intersection relationships between citation instances and intent classes, we map the fused representation $h_{i_{\text{fusion}}}^t$ into a box embedding space, as illustrated in Fig. 4(c). Unlike vector projections that fix a single position, this mapping constructs a spatial region for each citation. In this space, box volume reflects semantic breadth, and spatial overlap indicates class ambiguity.

Each citation instance is geometrically represented as a hyper-rectangle (box), and each intent class is parameterized by a learnable class box that is optimized jointly with the rest of the model during training. As shown in Eq. (12), each box is defined by two boundary vectors. This formulation allows classification to be modeled through geometric interactions between instance boxes and class boxes.

$$\text{Box} = (e, E), \quad \text{where } e < E, \quad (12)$$

First, we project the fused representation $h_{i_{\text{fusion}}}^t \in \mathbb{R}^{2d}$ into the box parameter space $\mathbb{R}^{2l}$ via Eq. (13).

$$h_{i_{2l}}^t = \mathbf{W} h_{i_{\text{fusion}}}^t + \mathbf{b} \in \mathbb{R}^{2l}. \quad (13)$$

We employ a projection layer to map the fused representation into geometric parameters, splitting the output into translation and expansion components. The first l dimensions define the lower boundary $h_{i_e}^t$, while the remaining dimensions determine the box scale δ via an exponential transform to enforce validity ($h_{i_e}^t > h_{i_e}^t$):

$$h_{i_{2l}}^t \in \mathbb{R}^{2l} \Rightarrow \begin{cases} h_{i_e}^t = h_{i_{2l}}^t[1:l], \\ \delta = \exp(h_{i_{2l}}^t[l+1:2l]) \Rightarrow (h_{i_e}^t, h_{i_E}^t) = h_{i_{\text{box}}}^t, \\ h_{i_E}^t = h_{i_e}^t + \delta, \end{cases} \quad (14)$$

The box embedding derived from the fused representation $h_{i_{\text{fusion}}}^t$ is defined in Eq. (15).

$$h_{i_{\text{box}}}^t = (h_{i_e}^t, h_{i_E}^t). \quad (15)$$

The volume of the box is calculated as the product of its edge lengths, defined in Eq. (16).

$$\text{Vol}(h_{i_{\text{box}}}^t) = \prod_{j=1}^l (h_{i_E, j}^t - h_{i_e, j}^t), \quad (16)$$

To prevent numerical underflow, we first compute the volumes logarithmically and subsequently recover the actual values through exponentiation. This process is outlined in Eqs. (17) and (18).

$$\log \text{Vol}(h_{i_{\text{box}}}^t) = \sum_{j=1}^l \log(h_{i_{E,j}}^t - h_{i_{E,j}}^t), \quad (17)$$

$$\text{Vol}(h_{i_{\text{box}}}^t) = \exp\left(\sum_{j=1}^l \log(h_{i_{E,j}}^t - h_{i_{E,j}}^t)\right). \quad (18)$$

here, j is the index of each coordinate dimension.

When the intersection of two box embeddings $h_{i_{\text{box}}}^t$ and $h_{j_{\text{box}}}^t$ is non-empty, it yields a new box $h_{i_{\text{intersect}_{\text{box}}}}^t = [h_{i_{\text{intersect}_e}}^t, h_{i_{\text{intersect}_E}}^t]$. The corresponding lower bound $h_{i_{\text{intersect}_e}}^t$ and upper bound $h_{i_{\text{intersect}_E}}^t$ are computed according to Eqs. (19) and (20).

$$h_{i_{\text{intersect}_e}}^t = \max(h_{i_e}^t, h_{j_e}^t), \quad (19)$$

$$h_{i_{\text{intersect}_E}}^t = \min(h_{i_E}^t, h_{j_E}^t). \quad (20)$$

If the intersection is empty (i.e., $h_{i_{\text{intersect}_e}}^t > h_{i_{\text{intersect}_E}}^t$ in any dimension), the intersection volume is defined as zero.

To quantify the relationship between a citation instance and a target class, we define the classification score as a probabilistic measure derived from geometric inclusion. Specifically, we use an **Intersection-over-Class (IoC)** formulation, rather than the Intersection over Union metric commonly used in object detection. The resulting score measures the proportion of the class box covered by its intersection with the instance box and can be interpreted as a geometric approximation of the likelihood that the citation instance belongs to the target class, as defined in Eq. (21).

$$\text{score}_{y_i^{\text{box}}, t} = \frac{\text{Vol}(h_{i_{\text{box}}}^t \cap h_{j_k^{\text{box}}}^t)}{\text{Vol}(h_{j_k^{\text{box}}}^t)}. \quad (21)$$

3.6. Loss function

The box embedding of the citation context $h_{i_{\text{box}}}^t$ yields a vector of classification scores across all classes, as defined in Eq. (22).

$$\text{logits}_{i_{\text{box}}}^t = \left[\text{score}_{y_i^{\text{box}}, t} \right]_{y_k=0}^{|Y|-1}, \quad (22)$$

The one-hot representation of the ground-truth label is given by Eq. (23).

$$\text{onehot}(h_{i_{\text{box}}}^t)_k = \begin{cases} 1, & k = y_i^t \\ 0, & \text{otherwise} \end{cases} \quad (23)$$

The corresponding loss is formulated using Binary Cross-Entropy (BCE). Since the classification score derived from box intersection already lies in the range $[0, 1]$, it can be directly interpreted as a probabilistic estimate of class inclusion. BCE therefore encourages larger overlap with the correct class while suppressing overlap with incorrect classes. The instance-level loss is defined in Eq. (24).

$$\mathcal{L}_{\text{box}}^t = \text{BCE}(\text{logits}_{i_{\text{box}}}^t, \text{onehot}(y_i^t)). \quad (24)$$

Notably, Eq. (24) optimizes the classification likelihood across all classes rather than computing an aggregate Intersection over Union metric. The total box loss $\mathcal{L}_{\text{box}}^t$ over the dataset $\mathcal{D}^t$ is formulated in Eq. (25).

$$\mathcal{L}_{\text{box}}^t = \frac{1}{|\mathcal{D}^t|} \sum_{(x_i^t, y_i^t) \in \mathcal{D}^t} \mathcal{L}_{i_{\text{box}}}^t, \quad (25)$$

Similarly, the fused representation is fed into a linear classifier to produce the class score vector $\text{logits}_{i_{\text{vec}}}^t$. Each class score $\text{score}_{y_i^{\text{vec}}, t}$ is computed through a linear projection as defined in Eq. (26).

$$\text{score}_{y_i^{\text{vec}}, t} = \mathbf{W} \cdot h_{i_{\text{fusion}}}^t + \mathbf{b}, \quad (26)$$

Table 1

Number and proportion of classes in the citation intent datasets.

Dataset	Citation Intent Class	Number of Instances	Distribution
ACL-ARC	background	997	51.37%
	motivation	88	4.53%
	uses	364	18.75%
	extends	72	3.71%
	comparison or contrast	351	18.08%
	future	69	3.55%
SciCite	Background information	6376	57.86%
	Method	3153	28.61%
	Result comparison	1491	13.53%

Table 2

Training, validation, and test set splits of the citation intent dataset.

Dataset	Train	Test	Validation
ACL-ARC	1529	139	273
SciCite	7784	1861	1375

Let $\text{logits}_{i_{\text{vec}}}^t \in \mathbb{R}^K$ denote the score vector of $h_{i_{\text{fusion}}}^t$ over K classes, with the loss function defined in Eq. (27).

$$\mathcal{L}_{i_{\text{vec}}}^t = \text{CE}(\text{logits}_{i_{\text{vec}}}^t, y_i^t). \quad (27)$$

The total vector loss $\mathcal{L}_{\text{vec}}^t$ over the dataset $\mathcal{D}^t$ is formulated in Eq. (28).

$$\mathcal{L}_{\text{vec}}^t = \frac{1}{|\mathcal{D}^t|} \sum_{(x_i^t, y_i^t) \in \mathcal{D}^t} \mathcal{L}_{i_{\text{vec}}}^t, \quad (28)$$

The final loss combines the vector-space loss and the box-space loss through a learnable weighting mechanism. Specifically, we introduce a trainable scalar parameter α' , and the effective weighting coefficient α is obtained by applying a sigmoid transformation so that $\alpha \in (0, 1)$:

$$\alpha = \sigma(\alpha') \quad (29)$$

The final joint loss is then defined as:

$$\mathcal{L}_{\text{total}}^t = \alpha \cdot \mathcal{L}_{\text{vec}}^t + (1 - \alpha) \cdot \mathcal{L}_{\text{box}}^t. \quad (30)$$

4. Experiments

4.1. Datasets

We evaluate our model on two standard citation intent benchmarks, ACL-ARC (Jurgens et al., 2018) and SciCite (Cohan et al., 2019), using the preprocessed versions from (Shui et al., 2024). Dataset statistics are summarized in Table 1 for class distributions and Table 2 for the train, validation, and test splits.

The ACL-ARC dataset comprises 186 NLP papers from the ACL anthology with 1941 labeled citation contexts. Each context is annotated with one of six intent classes: Background, Motivation, Use, Extends, Compare/Contrast, and Future.

The SciCite dataset includes 6627 computer science and medicine papers from the Semantic Scholar Open Research Corpus (Lo et al., 2019), featuring 11,020 labeled citation contexts across three intent classes: Background, Method, and Result.

4.2. Comparison methods

ImpactCite (Mercier et al., 2020) is based on the XLNet architecture and reports one of the strongest F1 scores among prior methods. We use the score reported in (Shui et al., 2024).

VarMAE (Hu et al., 2022) uses a domain-adaptive language model to map raw embeddings into a latent space through a contextual uncertainty learning module.

Table 3
Hyperparameter search space and selected values.

Parameter	Search Space	Selected Value
Batch Size	{8, 16, 32}	16
Learning Rate	{1e-5, 2e-5, 3e-5, 5e-5}	2e-5
Max Sequence Length	{128, 256, 512}	256
Training Epochs	{5, 10, 20}	10
Dropout Rate	{0.1, 0.2, 0.3}	0.1
Weight Decay	{0.0, 0.01, 0.1}	0.01

Table 4
Analysis of model stability across different random seeds (F1 score).

Seed	Seed 1	Seed 2	Seed 3	Seed 4	Seed 5	Mean $\pm$ Std
SciCite	87.22	87.60	87.74	87.97	87.92	87.69 $\pm$ 0.30
ACL-ARC	78.74	79.54	80.14	80.59	79.94	79.79 $\pm$ 0.69

We include CitePrompt (Lahiri et al., 2023) and MTL (Shui et al., 2024) as SciBERT-based baselines. CitePrompt fine-tunes SciBERT with a cue-learning strategy, whereas MTL uses a multi-dataset training strategy to improve performance on the target dataset and was the best citation intent classification model in 2024.

The large language model (LLM) series employs two approaches: (1) in-context learning and supervised fine-tuning (Koloveas et al., 2025) and (2) Incremental Contextual Curriculum Learning (ICCL) (Liu et al., 2024), which progressively increases example difficulty. The first approach was applied to multiple mainstream LLMs on the citation intent classification dataset.

4.3. Experiment settings and evaluation metrics

4.3.1. Implementation details

Our model is implemented in PyTorch with the Hugging Face Transformers library. All experiments were conducted on a single NVIDIA A100 GPU. To improve reproducibility and assess model stability, we performed grid search for hyperparameter selection based on validation performance. Table 3 summarizes the search space and the selected values.

For the weighting coefficient α , we introduce a trainable scalar parameter α' and compute the effective coefficient through a sigmoid transformation, i.e., $\alpha = \sigma(\alpha')$. This design ensures that the weight remains in the interval (0, 1) while allowing it to be optimized jointly with the other model parameters. We initialize α' to 0, which corresponds to an initial coefficient of $\alpha = 0.5$. To evaluate the robustness of GLB-Cite, we report the mean and standard deviation of the F1 scores over five random seeds (42, 123, 1, 77, and 2025), as shown in Table 4.

4.3.2. Evaluation metrics

We adopt precision (P), recall (R), F1 score, and accuracy (ACC) as evaluation metrics. For comparative analysis with prior work, we primarily report F1 scores due to their balanced consideration of precision and recall, as well as their consistency with existing literature. For comprehensive model analysis, we additionally provide P, R, and ACC metrics to thoroughly assess the contribution of each module.

4.4. Experimental results and analysis

4.4.1. Comparative analysis

The main results of the performance comparison between our GLB-Cite model and classical methods are shown in Table 5. Our method achieves state-of-the-art performance on both datasets, outperforming all baselines by significant margins: (+3.29% over VarMAE on ACL-ARC; +2.34% over MTL (2024) on SciCite). These consistent gains demonstrate our approach's superior ability to extract discriminative textual features for citation intent classification compared to direct SciBERT encoding. Notably, we observe even larger advantages over

Table 5
F1 scores (%) of different methods on the SciCite and ACL-ARC datasets.

Model Family	Method	SciCite	ACL-ARC
–	Feature-rich RF	–	53.00
	Structural Scaffolds	84.00	67.90
–	VarMAE	86.32	<u>76.50</u>
	Scaffolds	84.00	67.90
RNN	ICL 0-shot	72.86	38.55
GPT4	ICL 5-shot	74.55	50.18
Mixtral-8x7B-Instruct-v0.1	ICCL	66.76	–
Llama2-70B-Chat	ICCL	62.56	–
QWEN1.5-72B-CHAT	ICCL	75.02	–
Qwen2.5-14B	Fine-tuned Q8	86.47	68.48
	Fine-tuned FP16	86.84	67.73
XLNet	ImpactCite	84.98	64.62
BERT	Default	83.46	57.44
	MTL-TRL(BERT)	83.48	66.32
SciBERT	Default	85.22	67.25
	SciBERT + SPECTEP	–	64.15
	CitePrompt	85.02	66.58
	MTL-TRL	85.35	75.57
	GLB-Cite	87.69	79.79

Table 6
Impact of different modules on overall F1, precision, recall, and accuracy.

Dataset	Ablation Module	F1	P	R	ACC
SciCite	f + b	0.8769	0.8763	0.8801	0.8872
	with f	0.8727	0.8800	0.8693	0.8839
	with b	0.8686	0.8585	0.8808	0.8812
	with none	0.8471	0.8414	0.8535	0.8598
ACL-ARC	f + b	0.7979	0.8250	0.7798	0.8273
	with f	0.7864	0.8223	0.7712	0.8273
	with b	0.7920	0.8292	0.7700	0.8489
	with none	0.7587	0.7782	0.7521	0.8345

Qwen2.5-14B (+0.85% vs FP16-Qwen2.5-14B on SciCite; +11.31% vs Q8-Qwen2.5-14B on ACL-ARC), revealing the limitations of context-only fine-tuning and highlighting the need for specialized feature extraction architectures like ours. The main comparative results are reported in Table 5. GLB-Cite achieves state-of-the-art performance on both datasets, outperforming VarMAE by 3.29% on ACL-ARC and MTL (2024) by 2.34% on SciCite. These consistent gains indicate that GLB-Cite learns more discriminative representations than direct SciBERT encoding for citation intent classification. GLB-Cite also outperforms Qwen2.5-14B, with gains of 0.85% over FP16-Qwen2.5-14B on SciCite and 11.31% over Q8-Qwen2.5-14B on ACL-ARC. This result further highlights the limitations of context-only fine-tuning and the value of specialized architectures for feature extraction.

4.4.2. Module ablation study

To systematically examine when box embeddings are beneficial, we distinguish between two types of gains. The first is the standalone gain of box geometry over the plain encoder, defined as $\Delta_{\text{box}|\text{base}} = F1_b - F1_{\text{none}}$. The second is the marginal gain of BoxAlign on top of GL-Fusion, defined as $\Delta_{\text{box}|\text{fusion}} = F1_{f+b} - F1_f$. This distinction avoids the use of mixed baselines and makes the role of box embeddings easier to interpret at both the dataset level and the class level.

The results are shown in Table 6. Using both modules together yields the best overall performance. BoxAlign provides larger gains on ACL-ARC, which suggests that it better captures inter-class boundary relationships through geometric constraints. GL-Fusion contributes more on SciCite, highlighting the importance of fine-grained contextual semantic modeling. Their synergy balances local citation context with global class structure, ultimately enhancing classification accuracy.

To thoroughly assess each module's impact across classes, we evaluate performance at both dataset-wide and per-class levels, the results are shown in Tables 7 and 8.

Table 7
Ablation study results of classification on the SciCite dataset.

Class	Ablation Module	F1	P	R	ACC
Background	f + b	0.9040	0.8878	0.9208	0.9208
	with f	0.9027	0.8727	0.9348	0.9348
	with b	0.8952	0.9082	0.8826	0.8826
	with none	0.8775	0.8824	0.8726	0.8726
Method	f + b	0.8733	0.9196	0.8314	0.8314
	with f	0.8655	0.9314	0.8083	0.8083
	with b	0.8830	0.8867	0.8793	0.8793
	with none	0.8519	0.8576	0.8463	0.8463
Result	f + b	0.8534	0.8214	0.8880	0.8880
	with f	0.8501	0.8358	0.8649	0.8649
	with b	0.8276	0.7808	0.8803	0.8803
	with none	0.8119	0.7842	0.8417	0.8417

Table 8
Ablation study results of classification on the ACL-ARC dataset.

Class	Ablation Module	F1	P	R	ACC
Background	f + b	0.8725	0.8333	0.9155	0.9155
	with f	0.8701	0.8072	0.9437	0.9437
	with b	0.9007	0.8500	0.9577	0.9577
	with none	0.9020	0.8415	0.9718	0.9718
Uses	f + b	0.8511	0.9524	0.7692	0.7692
	with f	0.8511	0.9524	0.7692	0.7692
	with b	0.8444	1.0000	0.7308	0.7308
	with none	0.8696	1.0000	0.7692	0.7692
CompareOrContrast	f + b	0.7083	0.7391	0.6800	0.6800
	with f	0.7143	0.8824	0.6000	0.6000
	with b	0.7755	0.7917	0.7600	0.7600
	with none	0.6818	0.7895	0.6000	0.6000
Extends	f + b	0.8889	1.0000	0.8000	0.8000
	with f	0.8889	1.0000	0.8000	0.8000
	with b	0.8889	1.0000	0.8000	0.8000
	with none	0.8000	0.8000	0.8000	0.8000
Motivation	f + b	0.6667	0.6250	0.7143	0.7143
	with f	0.6667	0.6250	0.7143	0.7143
	with b	0.6154	0.6667	0.5714	0.5714
	with none	0.5714	0.5714	0.5714	0.5714
Future	f + b	0.8000	0.8000	0.8000	0.8000
	with f	0.7273	0.6667	0.8000	0.8000
	with b	0.7273	0.6667	0.8000	0.8000
	with none	0.7273	0.6667	0.8000	0.8000

Table 9
Class-wise gains brought by the BoxAlign module under two comparison settings.

Dataset	Class	$\Delta_{\text{box base}}$	$\Delta_{\text{box fusion}}$
SciCite	Background	+ 1.77	+ 0.13
	Method	+ 3.11	+ 0.78
	Result	+ 1.57	+ 0.33
ACL-ARC	Background	-0.13	+ 0.24
	Uses	-2.52	0.00
	CompareOrContrast	+ 9.37	-0.60
	Extends	+ 8.89	0.00
	Motivation	+ 4.40	0.00
	Future	0.00	+ 7.27

The experimental results in [Tables 7](#) demonstrate that incorporating information from the entire sentence leads to consistent improvements across all metrics compared to relying solely on the embedding at the citation marker. This enhancement is particularly pronounced in the F1 score, indicating that while the citation marker’s embedding plays a critical role in intent classification, the global context of the citation also contributes significantly to model performance. These findings suggest that an optimal approach should balance local marker features with broader contextual information [Table 9](#).

As shown in [Table 7](#), the introduction of the fusion module consistently improves performance across most metrics on the SciCite dataset compared to using only the citation marker representation. This demonstrates that effectively combining information from the entire sentence with localized citation marker information is essential for accurate citation intent classification. Relying solely on the citation marker representation fails to capture the full semantic context of the sentence.

At the dataset level, box embeddings improve the overall F1 score on both benchmarks, but their effect is more pronounced on ACL-ARC than on SciCite. Specifically, $\Delta_{\text{box|base}}$ reaches 3.33 points on ACL-ARC, compared with 2.15 points on SciCite, while $\Delta_{\text{box|fusion}}$ is 1.15 and 0.42 points, respectively. This difference is consistent with the characteristics of the two datasets. SciCite contains 11,020 citation contexts and adopts a coarse three-class taxonomy, whereas ACL-ARC contains 1941 citation contexts and involves six fine-grained intent classes. As a result, ACL-ARC presents more severe boundary ambiguity, making geometric region modeling more useful, but also more heterogeneous across classes.

The class-wise results further show that the benefit of box embeddings is not uniform. On SciCite, box embeddings provide stable improvements for all three classes, with the largest gain observed for Method, which improves by 3.11 points under $\Delta_{\text{box|base}}$ and 0.78 points under $\Delta_{\text{box|fusion}}$. Result also shows consistent positive gains under both comparisons. These findings suggest that box embeddings are particularly useful when a citation intent shares semantic cues with neighboring classes and cannot be fully separated by point-wise boundaries alone.

On ACL-ARC, the effect is more conditional. CompareOrContrast, Extends, and Motivation obtain clear gains under $\Delta_{\text{box|base}}$, indicating that box geometry is helpful when class regions overlap or partially contain one another. However, these gains are not always additive after GL-Fusion is introduced. For example, CompareOrContrast improves by 9.37 points relative to the plain encoder but decreases by 0.60 points relative to the fusion-only setting. This suggests that GL-Fusion and BoxAlign are partly complementary and partly overlapping in their contribution to class separation. A different pattern appears in Future, where box embeddings show no standalone gain but yield a 7.27-point improvement after fused context is introduced. This indicates that for some classes, geometric boundary modeling becomes effective only after the local and global cues have been sufficiently integrated.

Overall, these results suggest that box embeddings should be understood as a boundary refinement mechanism rather than a universal performance booster. They are most effective for classes with overlapping semantics and diffuse contextual cues. Their gains become limited when class boundaries are already well captured by contextual fusion or when the available training evidence is relatively sparse. These class-wise results also suggest that the effectiveness of box embeddings is not determined by class frequency alone. On SciCite, the largest gain is observed for *Method*, although it is not the smallest class. This pattern indicates that semantic overlap, rather than data scarcity alone, is the primary factor driving the usefulness of box embeddings. A similar pattern appears on ACL-ARC, where *CompareOrContrast* shows the largest standalone gain despite having more training instances than *Extends*, *Motivation*, and *Future*. At the same time, the unstable or non-additive gains observed for several low-resource classes suggest that limited training evidence can affect the stability of geometric region estimation. Overall, these results indicate that semantic boundary ambiguity is the main reason why box embeddings help, whereas class frequency mainly influences how consistently this benefit can be realized.

The experimental results on ACL-ARC indicate that the effects of GL-Fusion and BoxAlign are not identical across classes. In terms of overall F1, BoxAlign alone slightly outperforms GL-Fusion alone, while their combination achieves the best result. This pattern suggests that contextual fusion and geometric boundary modeling capture different aspects of citation intent and should be viewed as complementary rather than interchangeable.

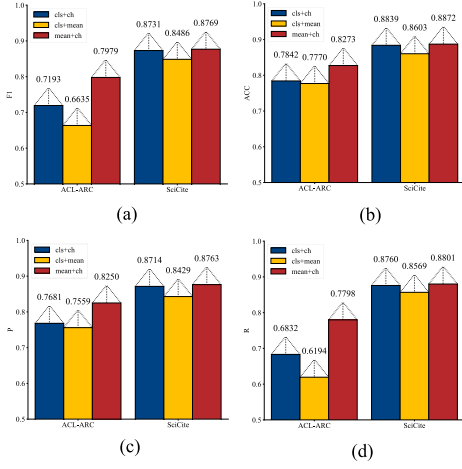


Fig. 5. Impact of different combination strategies.

Class-specific analysis reveals that the gains of box embeddings are highly class-dependent. The largest standalone improvements are observed for *CompareOrContrast* and *Extends*, while *Future* shows the clearest marginal gain after GL-Fusion is introduced. An additional pattern appears in *CompareOrContrast*: although box embeddings alone provide a substantial improvement, their gain is not further increased after being combined with GL-Fusion.

Overall, the integrated approach combining feature fusion with box embeddings demonstrates enhanced discriminative capability, especially for classes with overlapping semantics or ambiguous boundaries. At the same time, the class-wise results confirm that the contribution of box embeddings is conditional rather than uniform, which further highlights the importance of analyzing their effect at a finer granularity.

4.4.3. Component-wise evaluation of GL-Fusion

We compared feature combination methods (cls, mean, and ch) in Fig. 5 and observed clear dataset-dependent differences. On ACL-ARC, mean+ch improves the F1 score by 7.86% over cls+ch and by 13.44% over cls+mean. On SciCite, the gains are smaller, at 0.38% and 2.83%, respectively. These results show that ACL-ARC is more sensitive to feature combination strategies than SciCite.

Across both datasets, the mean+ch combination consistently achieves higher F1 scores than the alternative combinations and also outperforms the standalone ch setting. These results indicate that although localized citation marker information (ch) provides important discriminative cues, optimal performance requires complementary integration with a global contextual representation obtained through mean pooling. The consistent advantage of mean+ch therefore highlights the importance of combining both local and global information in citation intent classification.

The mean pooling strategy consistently performs better than cls because it aggregates information across all sentence tokens and therefore provides a more comprehensive global representation. By considering the contribution of each token, mean pooling captures the overall semantics of the citation context more effectively. This property is particularly beneficial for tasks that require robust global contextual modeling.

We also compared naive concatenation (cat) with our fusion approach (f), as shown in Figs. 6 and 7. Two findings emerge. First, ACL-ARC exhibits substantially greater performance variation across fusion methods than SciCite, indicating stronger sensitivity to feature integration strategies. Second, our fusion approach (f) consistently outperforms naive concatenation on both datasets, although the magnitude of improvement differs across datasets.

Our fusion method outperforms naive concatenation by integrating global and local features more effectively, yielding an 8.19% improve-

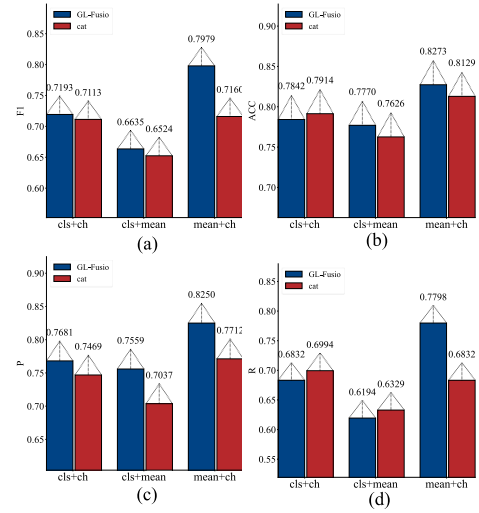


Fig. 6. Impact of different fusion strategies on the ACL-ARC dataset.

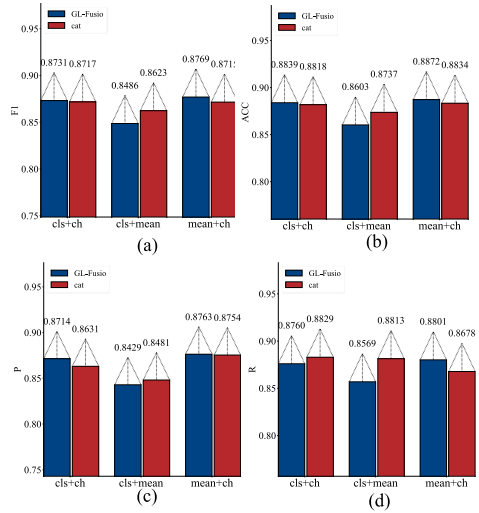


Fig. 7. Impact of different fusion strategies on the SciCite dataset.

ment in the mean+ch setting on ACL-ARC. Unlike naive concatenation, which may introduce feature interference, our approach adaptively balances feature importance, models feature interactions, and improves information integration. These properties are especially important for citation intent classification, which requires both contextual coverage and local precision.

4.4.4. Visualization analysis

To evaluate the effectiveness of our method, we visualize citation representations with t-SNE in Figs. 8 and 9. The left panels show the baseline representations, whereas the right panels show the representations learned by our method. In both datasets, the learned representations exhibit clearer inter-class separation, especially in previously ambiguous regions, indicating that our method produces more discriminative features.

Fig. 8 shows clearer class separation after applying our method. In the baseline visualization (left), *Background* and *Method* overlap substantially, *Result* samples are relatively dispersed, and inter-class mixing is common. In the improved visualization (right), the overlap between *Background* and *Method* is visibly reduced, *Result* samples form a more compact cluster, and the class boundaries become clearer

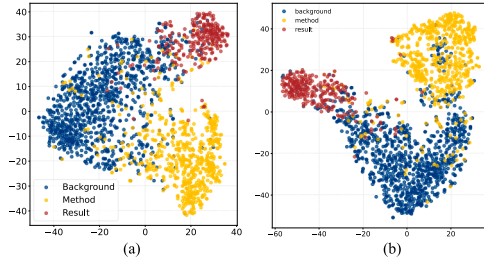


Fig. 8. t-SNE visualization of the SciCite dataset.

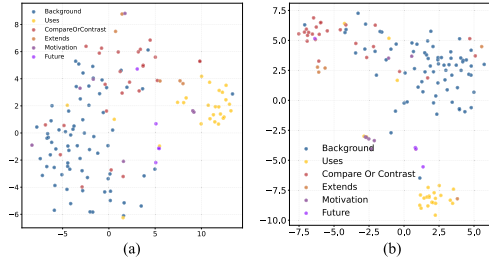


Fig. 9. t-SNE visualization of the ACL-ARC dataset.

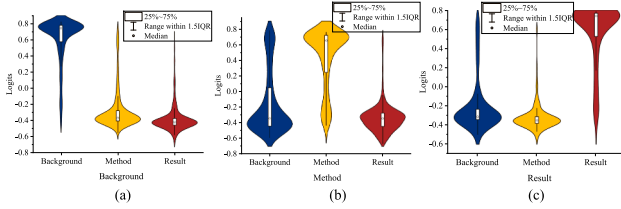


Fig. 10. Pre-improvement class-conditional logit distributions on the SciCite training set.

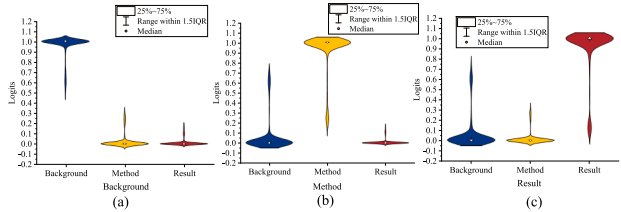


Fig. 11. Post-improvement class-conditional logit distributions on the SciCite training set.

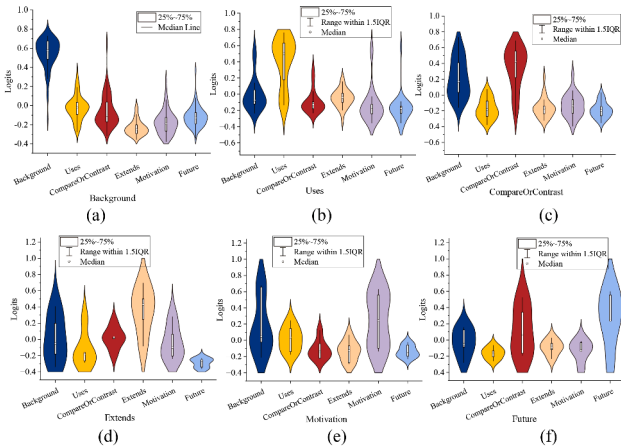


Fig. 12. Pre-improvement class-conditional logit distributions on the ACL-ARC training set.

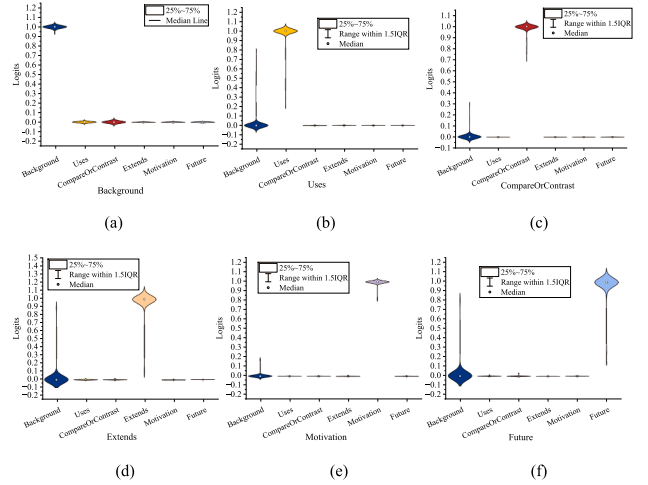


Fig. 13. Post-improvement class-conditional logit distributions on the ACL-ARC training set.

overall. These patterns suggest that our method learns more discriminative representations.

Fig. 9 shows clearer clustering patterns on ACL-ARC after applying our method. In the baseline visualization (left), *CompareOrContrast* samples are relatively diffuse and *Motivation* samples appear scattered. In the improved visualization (right), *CompareOrContrast* forms a more coherent cluster and *Motivation* samples become more tightly grouped. These visual changes further support the ability of our method to learn more discriminative representations for fine-grained citation intent classification.

To examine how feature fusion and geometric modeling affect classification separability, we analyze class-conditional logit distributions for correct and incorrect predictions across classes.

Figs. 10–13 show two notable changes after applying our method: (1) clearer separation between class-conditional logit distributions, and (2) generally higher logits for correct class predictions. The reduced overlap between correct and incorrect distributions indicates improved discriminative power compared with the baseline.

5. Conclusion

We propose GLB-Cite, a dual-embedding framework for citation intent classification. The framework addresses inadequate context modeling, ambiguous class boundaries, and limited representation capacity through three key components: GL-Fusion, box embeddings, and a joint optimization strategy for vector-space and box-space learning. Experiments on SciCite and ACL-ARC achieve state-of-the-art F1 scores of 87.69% and 79.79%, respectively, while also improving robustness to class overlap and semantic ambiguity. These results demonstrate the value of geometric learning for citation analysis in scientometrics.

Although the proposed method yields substantial performance gains, its F1 score on ACL-ARC remains below 80%. We hypothesize that this limitation is partly due to the representational constraints of SciBERT. Future work will explore stronger scientific text encoders, more expressive geometric representations, and more adaptive mechanisms for boundary modeling.

CRedit authorship contribution statement

Jinwen Yang: Writing – original draft, Methodology, Investigation, Data Curation and Software, Formal analysis, Conceptualization; **Zhi-juan Du:** Conceptualization, Methodology, Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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